

DATABASE SYSTEMS ENGG AND DISTRIBUTED BACKEND DEVELOPMENT

Week 2:

```
-> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM books
-> ;
+-----+-----+-----+-----+
| book_id | title                | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby     | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984                 | 9780451524935 | 1949           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Members (
->     member_id INT PRIMARY KEY,
->     full_name VARCHAR(100),
->     email VARCHAR(100) UNIQUE
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Members (member_id, full_name, email)
-> VALUES
-> (101, 'John Smith', 'john.smith@email.com'),
-> (102, 'Emma Wilson', 'emma.wilson@email.com'),
-> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> create database bookflow3_db
-> ;
Query OK, 1 row affected (0.03 sec)

mysql> USE bookflow3_db
Database changed
mysql> USE bookflow3_db
Database changed
mysql> CREATE TABLE Books(
->     book_id INT PRIMARY KEY,
->     title VARCHAR(100) NOT NULL,
->     isbn VARCHAR(20) UNIQUE,
->     published_year INT CHECK(published_year<2027)
-> );
Query OK, 0 rows affected (0.09 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES
-> (1, 'The Great Gatsby', '9780743273565', 1925),
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
-> (3, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM books
-> ;
+-----+-----+-----+-----+
| book_id | title                | isbn          | published_year |
+-----+-----+-----+-----+
```

Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Loans;

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

10 rows in set (0.00 sec)

mysql> SELECT

-> m.full_name AS Member_Name,
-> b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m
-> ON l.member_id = m.member_id
-> INNER JOIN Books b
-> ON l.book_id = b.book_id;

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby
Emma Wilson	1984
Michael Brown	1984
Michael Brown	To Kill a Mockingbird
Michael Brown	The Great Gatsby

mysql> SELECT * FROM Members

-> ;

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

3 rows in set (0.00 sec)

mysql> CREATE TABLE Loans (

-> loan_id INT PRIMARY KEY,
-> member_id INT,
-> book_id INT,
-> loan_date DATE,
-> FOREIGN KEY (member_id) REFERENCES Members(member_id),
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
->);

Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date)

-> VALUES

-> (1, 101, 1, '2025-01-05'),
-> (2, 102, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'),
-> (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'),
-> (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'),

10 rows in set (0.02 sec)

```
mysql> SELECT
->     published_year,
->     COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1

3 rows in set (0.02 sec)

```
mysql> CREATE TABLE Donation_History (
->     donation_id INT PRIMARY KEY,
->     book_id INT,
->     donor_name VARCHAR(100),
->     donation_date DATE,
->     FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO Books
-> (book_id, title, isbn, published_year)
-> VALUES
-> (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO Donation_History
-> (donation_id, book_id, donor_name, donation_date)
-> VALUES
-> (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)
```

```
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
|          1 |        4 | Raj Kumar  | 2026-08-01    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> CREATE INDEX idx_books_isbn
-> ON Books(isbn);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT *
-> FROM Books
-> WHERE isbn = '9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn          | published_year |
+-----+-----+-----+-----+
|        3 | 1984 | 9780451524935 |          1949 |
+-----+-----+-----+-----+
1 row in set (0.02 sec)
```

```
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title                | isbn          | published_year |
+-----+-----+-----+-----+
|        1 | The Great Gatsby     | 9780743273565 |          1925 |
|        2 | To Kill a Mockingbird | 9780061120084 |          1960 |
|        3 | 1984                 | 9780451524935 |          1949 |
|        4 | Animal Farm          | 9780451526342 |          1945 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Members;
+-----+-----+-----+-----+
|
```

1 row in set (0.02 sec)

mysql> SELECT * FROM Books;

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
4	Animal Farm	9780451526342	1945

4 rows in set (0.00 sec)

mysql> SELECT * FROM Members;

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

3 rows in set (0.00 sec)

mysql> SELECT * FROM Loans;

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
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5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

10 rows in set (0.00 sec)

mysql> SELECT * FROM Donation_History;

25°C
Partly sunny



Search

